

1. Given that the link rate $L = 4\text{Mbps} = 4 \times 10^6$ bps. Since there are two packet transmitting $R = 1500$ bytes $= 1500 * 8 = 12000$ bits from the hosts, the transmission delay is

$$\frac{R}{L} = \frac{12000}{4 \times 10^6} = \frac{3}{1000} = 3ms$$

The transmission delay refers to the time taken for the last packet about to be transmitted to the end router. The propagation delay refers to the time taken for one packet to be transmitted through the link. Given that the transmission delay of the both packets is 3ms, one of the packets needs 6ms to be transmitted while another needs 2ms.

Therefore, one of the hosts will send the last packet to the router A at 5ms, and the another host will send the first packet at 6ms. Since the nodal processing time is negligible, the packet will be transmitted through the router once arrived. There will be no queuing delay.

2. Given that $F = 17Gbits$, $d_{min} = 4Mbps$ and $u_s = 40Mbps$ and so the mininum client download rate is $F/d_{min} = (17 \times 10^9)/(4 \times 10^6) = 4.25 \times 10^3 = 4250$.

Since the time taken for sending N copies is compputed as NF/u_s , and we have $F/u_s = (17 \times 10^9)/(40 \times 10^6) = 425$, then we have:

When $N = 20$, $NF/u_s = 425 * 20 = 8500$

When $N = 200$, $NF/u_s = 425 * 200 = 85000$

When $N = 2000$, $NF/u_s = 425 * 2000 = 850000$

Note that $D_{C-S} = \max \{NF/u_s, F/d_{min}\}$, the minimum distribution time of Client-server model:

	N = 20	N = 200	N = 200
u = 400kbps	8500	85000	850000
u = 600kbps	8500	85000	850000
u = 4Mbps	8500	85000	850000

As calculated, $F/u_s = (17 * 10^9)/(40 * 10^6) = 425$

$F/d_{min} = (17 \times 10^9)/(4 \times 10^6) = 4.25 \times 10^3 = 4250$

When $N = 20$, $u = 400kbps$, $(20 * 17 * 10^9)/(40 * 10^6 + 20 * 400 * 10^3) = 7083.333333$

When $N = 200$, $u = 400kbps$, $(200 * 17 * 10^9)/(40 * 10^6 + 200 * 400 * 10^3) = 28333.333333$

When $N = 2000$, $u = 400kbps$, $(2000 * 17 * 10^9)/(40 * 10^6 + 2000 * 400 * 10^3) = 40476.190476$

When $N = 20$, $u = 600kbps$, $(20 * 17 * 10^9)/(40 * 10^6 + 20 * 600 * 10^3) = 6538.461538$

When $N = 200$, $u = 600kbps$, $(200 * 17 * 10^9)/(40 * 10^6 + 200 * 600 * 10^3) = 21250$

When $N = 2000$, $u = 600kbps$, $(2000 * 17 * 10^9)/(40 * 10^6 + 2000 * 600 * 10^3) = 27419.354839$

When $N = 20$, $u = 4Mbps$, $(20 * 17 * 10^9)/(40 * 10^6 + 20 * 4 * 10^6) = 2833.333333$

When $N = 200$, $u = 4Mbps$, $(200 * 17 * 10^9)/(40 * 10^6 + 200 * 4 * 10^6) = 4047.619048$

When $N = 2000$, $u = 4Mbps$, $(2000 * 17 * 10^9)/(40 * 10^6 + 2000 * 4 * 10^6) = 4228.855721$

Note that $D_{P2P} = \max \{F/u_s, F/d_{min}, NF/(u_s + \sum u_i)\} = \max \{F/u_s, F/d_{min}, NF/(u_s + Nu)\}$, the minimum distribution time of Client-server model: P2P distribution

	N = 20	N = 200	N = 200
u = 400kbps	7083	28333	40476
u = 600kbps	6538	21250	27419
u = 4Mbps	4250	4250	4250

3. (a) For the segments flowing from the server B to the clients' processes, we have

Server	Client	source port	destination port
B	A	80	26145
B	C: left process	80	7532
B	C: right process	80	26145

- (b) For the segments flowing from the server B to the clients' processes, we have

Server	Client	source IP	dest IP	source port	des port
B	A	b	a	80	26145
B	C: left process	b	c	80	7532
B	C: right process	b	c	80	26145

4. (a) We can look into the problem through different cases.

Case 1: Sender has received ACK value for all the packets inside the window, then the send base of the sender's window will be moved to k and hence the sender's window would be $[k, k+3]$;

Case 2: Sender has not received ACK value for all the packets inside the window, then the send base of the sender's window will not be moved, and hence the sender's window would be $[k-4, k-1]$.

Case 3: Sender has not received ACK value for some of the packets inside the window, then the send base will be moved somewhere to $[k-3, k-1]$, and hence the sender's window would be $[k-3, k]$, $[k-2, k+1]$, $[k-1, k+2]$.

- (b) As the receiver is expecting packet k , then it has received packets $k-4, k-3, k-2, k-1$ and acknowledged those packets. There may be some ACK value $k-4, k-3, k-2, k-1$ propagating back to the sender if not all of these ACK messages have received by the sender.

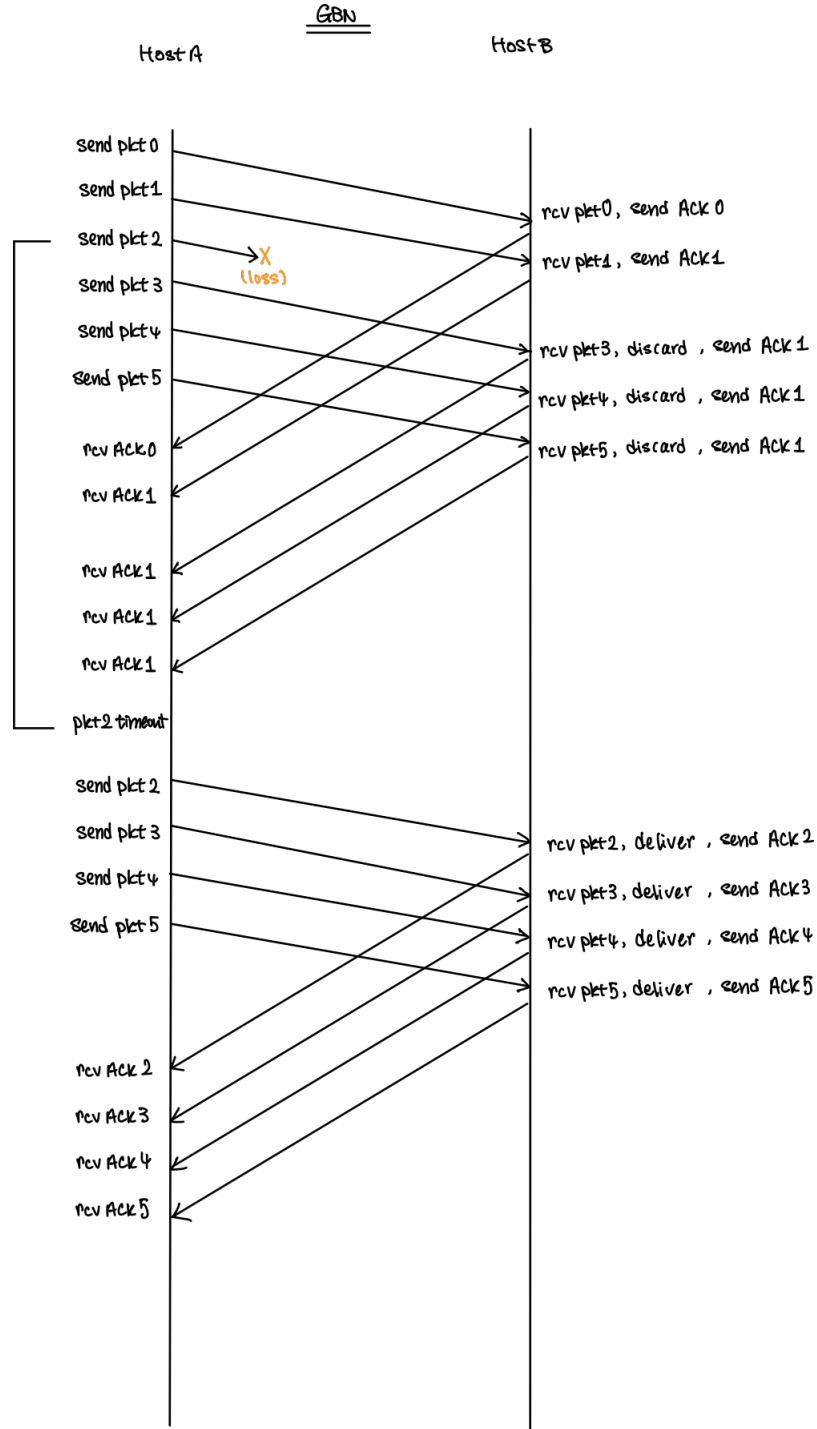
The sender must have already received an ACK value $k-5$ as the window starts from $k-4$, so the receiver will not send any ACK value less than $k-5$.

Therefore, the range of all possible values of the ACK messages currently propagating back to the sender at time t would be $[k-5, k-1]$.

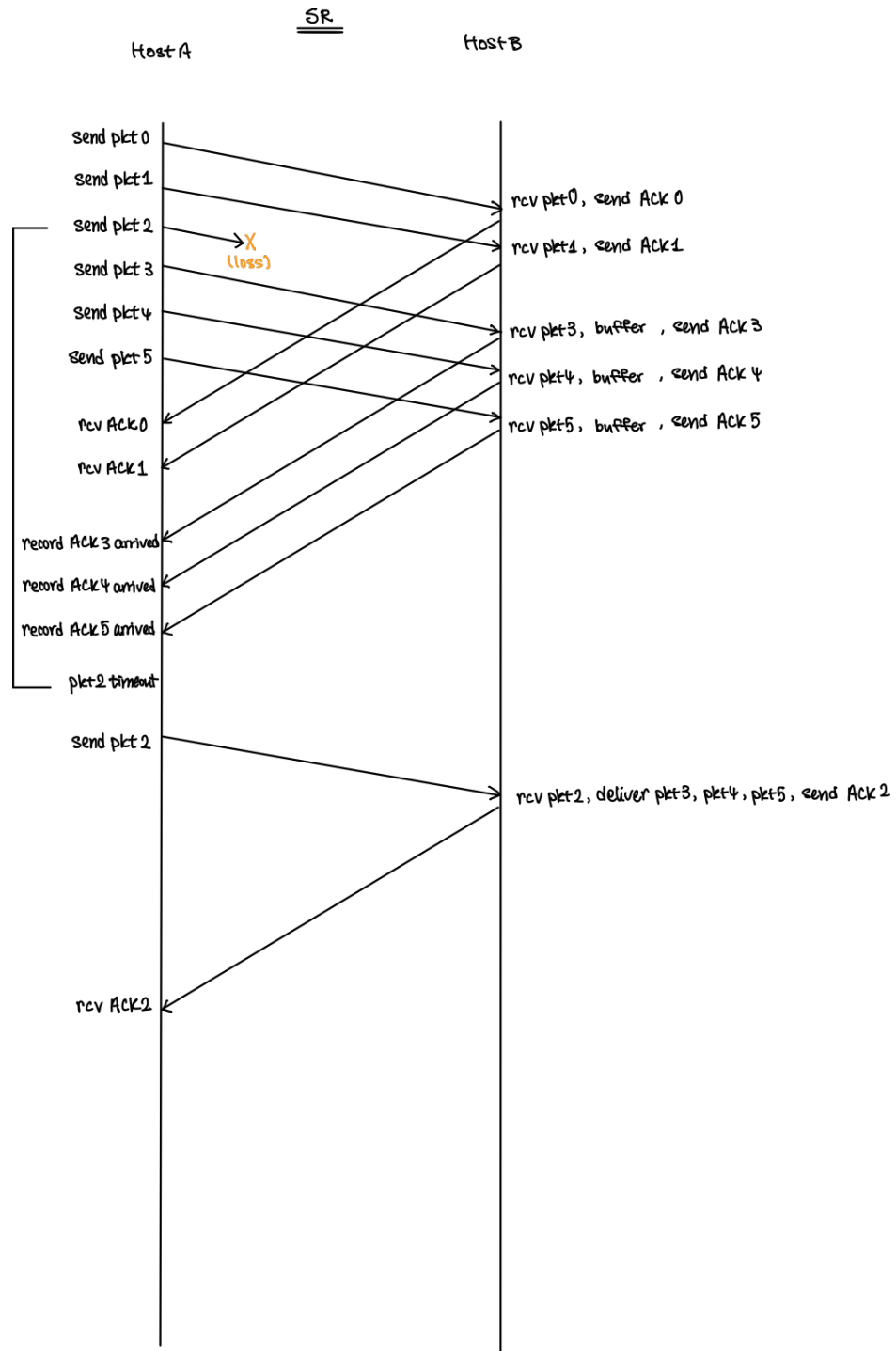
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- The diagram illustrates the Go-Back-N protocol between Host A and Host B. The sender window size is N=3.
- Host A (Sender):**
- Initial state: Buffer [0, 1, 2] | 3, 4, 5
 - After receiving ACK 1: Buffer [2, 3, 4] | 5
 - After receiving ACK 2: Buffer [3, 4, 5]
 - After receiving ACK 3: Buffer [4, 5]
 - After packet 4 timeout: Buffer [4, 5]
 - After retransmission: Buffer [5]
 - Final state: Buffer [5]
- Host B (Receiver):**
- Initial state: Empty
 - After receiving pkt 0: Buffer [0] | send ACK 0
 - After receiving pkt 1: Buffer [0, 1] | send ACK 1
 - After receiving pkt 2: Buffer [0, 1, 2] | send ACK 2
 - After receiving pkt 3: Buffer [0, 1, 2, 3] | deliver, send ACK 3
 - After receiving pkt 5: Buffer [0, 1, 2, 3, 5] | discard, send ACK 3
 - After receiving ACK 3: Buffer [0, 1, 2, 3, 5] | deliver, send ACK 4
 - After receiving pkt 4: Buffer [0, 1, 2, 3, 4, 5] | deliver, send ACK 5
 - After receiving pkt 5: Buffer [0, 1, 2, 3, 4, 5]
- Sequence of Events:**
- Host A sends pkt 0, 1, 2.
 - Host B receives pkt 0, 1, 2 and sends ACK 2.
 - Host A receives ACK 2 and sends pkt 3, 4, 5.
 - Host B receives pkt 3 and sends ACK 3.
 - Host B receives pkt 5 and sends ACK 3 (discards pkt 5).
 - Host A receives ACK 3 and sends pkt 4 (retransmission).
 - Host B receives pkt 4 and sends ACK 4.
 - Host A receives ACK 4 and sends pkt 5.
 - Host B receives pkt 5 and sends ACK 5.

6. (a) Assume that for 3 timing diagram GAN, SR and TCP belows, the sliding window size is much bigger than 6 so that we need not to consider the move of the sliding window.

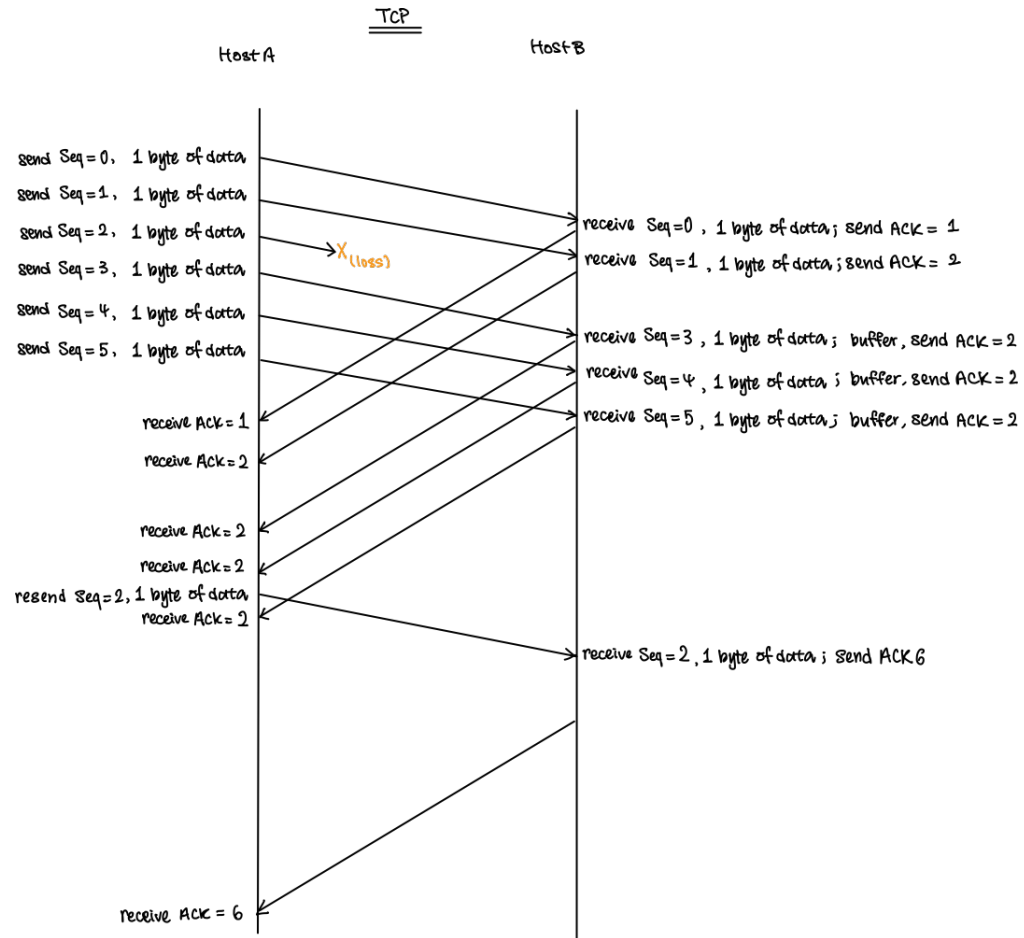
(i) The timing diagram between Host A and Host B for GBN:



(ii) The timing diagram between Host A and Host B for SR:



- (iii) Further assume that the 3-way handshake and closing are excluded, and the congestion control are not considered in this situation. The timing diagram between Host A and Host B for TCP:



- (b) (i) For GBN, Host A sends 10 segments in total. Segments 0, 1, 2, 3, 4, 5 are first sent, and segments 2, 3, 4, 5 is resent. Host B sends 9 segments in total. ACKs 0, 1, 1, 1, 1 are first sent, and ACKs 2, 3, 4, 5 are later sent.
- (ii) For SR, Host A sends 7 segments in total. Segments 0, 1, 2, 3, 4, 5 are first sent, and segments 2 is resent. Host B sends 6 segments in total. ACKs 0, 1, 3, 4, 5 are first sent, and ACKs 2 are later sent.
- (iii) For TCP, Host A sends 7 segments in total. Segments 0, 1, 2, 3, 4, 5 are first sent, and segments 2 is resent. Host B sends 6 segments in total. ACKs 1, 2, 2, 2, 2 are first sent, and ACKs 6 are later sent.